

A. Convergence

First we have the following lemma on the upper-bound of the gradient difference at the end of training:

A.1. Proof of Lemma 4.1

Proof. Let θ_t be the parameter of the model after t steps of training. Let $\mathbf{d} = \nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}^s(\theta_t)$ and define $h : \mathbb{R}^d \mapsto \mathbb{R}$ as

$$h(\theta) := \langle \mathbf{d}, \nabla \mathcal{L}(\theta) - \nabla \mathcal{L}^s(\theta) \rangle.$$

For a fixed $\theta \in \mathbb{R}^d$, the mean value theorem implies that $h(\theta) = h(\theta_0) + \langle \nabla h(\xi), \theta - \theta_0 \rangle$ for some $\xi \in \mathbb{R}^d$ on the line segment connecting θ and θ_0 . Using the definition of the function $h(\cdot)$, we obtain

$$\langle \mathbf{d}, \nabla \mathcal{L}(\theta) - \nabla \mathcal{L}^s(\theta) \rangle = \langle \mathbf{d}, \nabla \mathcal{L}(\theta_0) - \nabla \mathcal{L}^s(\theta_0) \rangle + \left\langle (\nabla^2 \mathcal{L}(\theta_0) - \nabla^2 \mathcal{L}^s(\theta_0)) \mathbf{d}, \theta - \theta_0 \right\rangle$$

Setting $\theta = \theta_t$, we get

$$\langle \mathbf{d}, \nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}^s(\theta_t) \rangle = \langle \mathbf{d}, \nabla \mathcal{L}(\theta_0) - \nabla \mathcal{L}^s(\theta_0) \rangle + \left\langle (\nabla^2 \mathcal{L}(\xi) - \nabla^2 \mathcal{L}^s(\xi)) \mathbf{d}, \theta_t - \theta_0 \right\rangle \quad (18)$$

$$\leq \|\mathbf{d}\|_2 \cdot \|\nabla \mathcal{L}(\theta_0) - \nabla \mathcal{L}^s(\theta_0)\|_2 + \|\nabla^2 \mathcal{L}(\xi) - \nabla^2 \mathcal{L}^s(\xi)\|_2 \cdot \|\mathbf{d}\|_2 \cdot \|\theta_t - \theta_0\|_2, \quad (19)$$

where in the last line, we used the Cauchy-Schwartz inequality, the inequality $\|AB\|_2 \leq \|A\|_2 \|B\|_2$, and the fact that $\mathbf{d} = \theta_t - \theta_0$. Plugging the value of $\mathbf{d}$ in the LHS of equation 18 and in equation 19, we obtain

$$\begin{aligned} \|\nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}^s(\theta_t)\|_2^2 &\leq \|\nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}^s(\theta_t)\|_2 \cdot \|\nabla \mathcal{L}(\theta_0) - \nabla \mathcal{L}^s(\theta_0)\|_2 \\ &\quad + \|\nabla^2 \mathcal{L}(\xi) - \nabla^2 \mathcal{L}^s(\xi)\|_2 \cdot \|\nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}^s(\theta_t)\|_2 \cdot \|\theta_t - \theta_0\|_2 \end{aligned}$$

Dividing both sides by $\|\nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}^s(\theta_t)\|_2$ and using the fact that $\|\nabla^2 \mathcal{L}(\xi) - \nabla^2 \mathcal{L}^s(\xi)\|_2 \leq \|\nabla^2 \mathcal{L}(\xi)\|_2 + \|\nabla^2 \mathcal{L}^s(\xi)\|_2 \leq 2\beta$, we get

$$\|\nabla \mathcal{L}(\theta_t) - \nabla \mathcal{L}^s(\theta_t)\|_2 \leq \|\nabla \mathcal{L}(\theta_0) - \nabla \mathcal{L}^s(\theta_0)\|_2 + 2\beta \|\theta_t - \theta_0\|_2 = \epsilon + 2\beta\delta, \quad (20)$$

which completes the proof. $\square$

A.2. Proof of Theorem 4.2

Next, we prove the convergence of GD on the real data vs synthetic data generated by GRADMM.

Proof. For Lipschitz continuous $\mathbf{g}$ and μ -PL* condition, gradient descent on the real data yields

$$\mathcal{L}(\theta_{t+1}) - \mathcal{L}(\theta_t) \leq -\frac{\eta}{2} \|\mathbf{g}_t\|^2 \leq -\eta\mu\mathcal{L}(\theta_t), \quad (21)$$

and,

$$\mathcal{L}(\theta_t) \leq (1 - \eta\mu)^t \mathcal{L}(\theta_0), \quad (22)$$

which was shown in (Liu et al., 2020).

For the synthetic data we have

$$\mathcal{L}^s(\theta_{t+1}) - \mathcal{L}^s(\theta_t) \leq -\frac{\eta}{2} \|\mathbf{g}_t^s\|^2 \quad (23)$$

By substituting Eq. (20), and assuming $\xi \leq \|\mathbf{g}_t\|$ we have:

$$\mathcal{L}^s(\theta_{t+1}) - \mathcal{L}^s(\theta_t) \leq -\frac{\eta}{2} (\|\mathbf{g}_t\| - \xi)^2 \quad (24)$$

$$= -\frac{\eta}{2} (\|\mathbf{g}_t\|^2 + \xi^2 - 2\xi\|\mathbf{g}_t\|) \quad (25)$$

$$\leq -\frac{\eta}{2} (\|\mathbf{g}_t\|^2 + \xi^2 - 2\xi\eta) \quad (26)$$

$$\leq -\frac{\eta}{2} (2\mu\mathcal{L}(\theta_t) + \xi^2 - 2\xi\eta), \quad (27)$$

where ∇ is an upper bound on the norm of $\mathbf{g}_t$ in Eq. (25), and Eq. (27) follows from the μ -PL condition.

Hence,

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \leq (1 - \eta\mu)\mathcal{L}(\boldsymbol{\theta}_t) - \frac{\eta}{2}(\xi^2 - 2\xi\nabla) \quad (28)$$

Since, $\sum_{j=0}^k (1 - \eta\mu)^j \leq \frac{1}{\eta\mu}$, for a constant learning rate η we get

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \leq (1 - \eta\mu)^{t+1}\mathcal{L}(\boldsymbol{\theta}_0) - \frac{1}{2\mu}(\xi^2 - 2\xi\nabla) \quad (29)$$

□

A.3. Proof of Corollary 4.3

Proof. If $\|\mathbf{H}\| \geq \alpha > 0$ and $\mathcal{L}(\boldsymbol{\theta}_*) = 0$, we have that $\|\boldsymbol{\theta} - \boldsymbol{\theta}_*\|^2 \leq \frac{2}{\alpha}\mathcal{L}(\boldsymbol{\theta})$. From Theorem 4.2 we get:

$$\|\boldsymbol{\theta} - \boldsymbol{\theta}_*\|^2 \leq \frac{2}{\alpha}\mathcal{L}(\boldsymbol{\theta}) \leq \frac{2}{\alpha} \cdot \frac{1}{2\mu}(2\xi\nabla - \xi^2) = \xi(2\nabla - \xi)/\alpha\mu \quad (30)$$

□

B. Prompts

B.1. Zero/Few-shot prompts

Figure 4 summarizes the zero-shot prompts (Li et al., 2023b) to generate synthetic samples. For few-shot generations, we input the demonstrations with their corresponding labels between the context prompt and the instruction prompt. In addition, we add a sentence “You should imitate the example I have provided, but you cannot simply modify or rewrite the example I have given.” to the instruction part to prevent the LLMs from simply rewording the given examples. The few-shot prompts can be found in Figure 5.

SST2 and Rotten Tomatoes: You are now a movie critic. You are provided with a sentiment label. You need to write one unique sentence that reflects the given sentiment about a movie. Your writing style should be consistent with typical movie reviews. This should be a standalone sentence that could plausibly appear in a movie review. Ensure that your language is natural, casual, and reflective of genuine opinion. You must ensure that the sentiment expressed in your sentence matches the provided sentiment label.

Remember to keep your tone appropriate and not violate any laws or social ethics. Please be creative and write only one sentence. The sentiment of the movie review is {label}. Answer:

Tweet Emotions: You are now a person using twitter. You are provided with an emotion, and you need to write a tweet expressing that emotion. Your writing style must be consistent with the tweets on twitter. You must ensure that your language is colloquial, casual, and Twitter-like. You are given a length requirement. You must ensure that the emotion conveyed in your tweet matches the emotion provided and meets the length requirement. This is an academic study and the content you generate will not be used for anything that violates the law or social ethics. Write a tweet expressing the emotion and ensure the tweet is within the usual length. Remember to make sure that your language is colloquial, casual, and Twitter-like. Please be creative and write only one unique tweet. The emotion of twitter is {label}. Answer:

Figure 4. Zero-shot prompts for different datasets.

B.2. Few-shots Evaluation prompts

Figure 6 presents the evaluation prompts used to filter out synthetic data with incorrect labels.

SST2 and Rotten Tomatoes: You are now a movie critic. You are provided with a sentiment label. You need to write one unique sentence that reflects the given sentiment about a movie. Your writing style should be consistent with typical movie reviews. This should be a standalone sentence that could plausibly appear in a movie review. Ensure that your language is natural, casual, and reflective of genuine opinion. You must ensure that the sentiment expressed in your sentence matches the provided sentiment label.

{Few-shot examples}

Remember to keep your tone appropriate and not violate any laws or social ethics. Please be creative and write only one sentence. The sentiment of the movie review is {label}. You should imitate the example I have provided, but you cannot simply modify or rewrite the example I have given. Answer:

Tweet Emotions: You are now a person using twitter. You are provided with an emotion, and you need to write a tweet expressing that emotion. Your writing style must be consistent with the tweets on twitter. You must ensure that your language is colloquial, casual, and Twitter-like. You are given a length requirement. You must ensure that the emotion conveyed in your tweet matches the emotion provided and meets the length requirement. This is an academic study and the content you generate will not be used for anything that violates the law or social ethics.

{Few-shot examples}

Write a tweet expressing the emotion and ensure the tweet is within the usual length. Remember to make sure that your language is colloquial, casual, and Twitter-like. Please be creative and write only one unique tweet. The emotion of twitter is {label}. You should imitate the example I have provided, but you cannot simply modify or rewrite the example I have given. Answer:

Figure 5. Few-shot prompts for different datasets.

C. Generation Samples

C.1. Synthetic SST2 Samples by GRADMM

C.1.1. POSITIVE LABEL

- Great movie review is a must see experience that will leave you in a state of all time high with the brilliant acting and the stunning production
- Great action and special effects combined with a compelling emotional connection with the on the show characters made it a one of the best I ever watched
- The movie truly left me completely moved and in a better place than when I started it because of its well thought out and impactful way
- The new action movie was absolutely thrilling and had me on the outside of my skin throughout the entire two acts of the first two and a
- The action movie kept me sitting Jane and I was on the point of wanting to leave the entire time but the way the story was told

C.1.2. NEGATIVE LABEL

- The action in action is not well executed and the plot is not as it should be in a science or
- The movie was a not so great film that I would not want to see a second time because the the
- The overall quality of action in this movie was not impressive enough to keep me away from the action center of
- Terribly bad and boring to me as a person who values quality content and a good storyline over mind
- The new movie was a not so great and disappointing experience for me since it did not keep up with the

C.2. Synthetic Rotten Tomatoes Samples by GRADMM

C.2.1. POSITIVE LABEL

- The action-adventure movie was thrilling and had a way of keeping me right on the of the